

CPEN 416 / EECE 5710

Lecture 14: QPE and order finding

Tuesday 28 October 2025

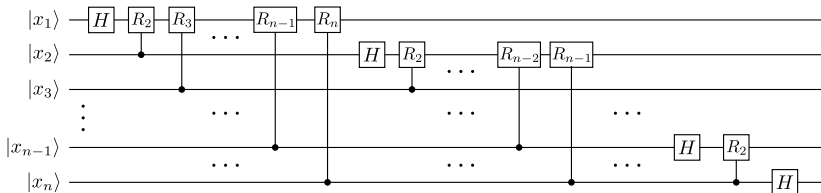
Announcements

- Quiz 6 today
- Tutorial Assignment 3 due Friday 23:59
- Midterm checkpoint report due Friday **5:00pm**
 - Calendar bookings for checkpoints meetings available Thursday after class
- Practice problem set 3 available (about QPE)

Last time

We implemented the quantum Fourier transform using a *polynomial* number of gates and depth.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$



Last time

We applied the QFT to quantum phase estimation (QPE) to estimate eigenvalues of unitary matrices.

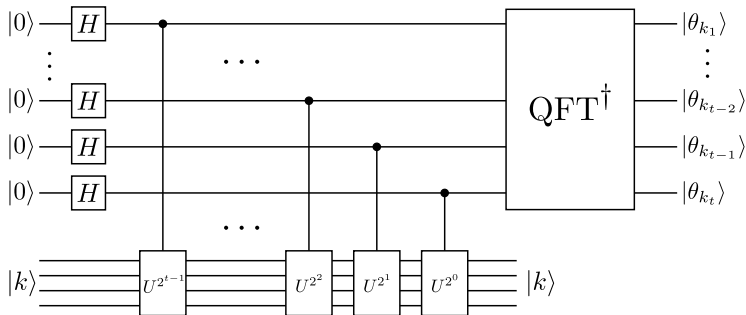
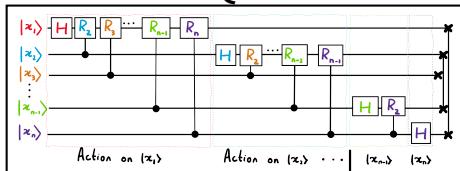


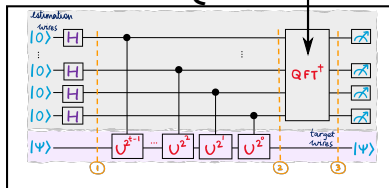
Image credit: Xanadu Quantum Codebook node P.2

Reminder: where are we going?

1. QFT



2. QPE



3. Shor

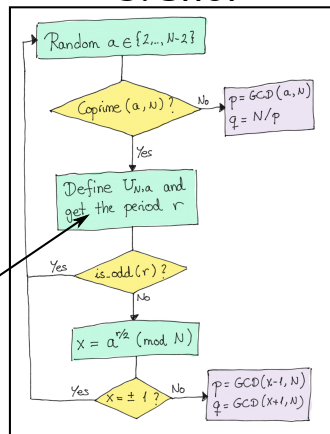


Image credit: Xanadu Quantum Codebook nodes F.3, P.3, S.4

Learning outcomes

- Perform arithmetic modulo N with integers and with quantum operations
- Define the order finding problem
- Apply QPE to solve the order finding problem

Modular arithmetic

Consider the finite set of integers $\{0, 1, \dots, N - 1\}$.

Basic arithmetic modulo N :

Modular inverses

The inverse of a modulo N (a^{-1}) is an integer b such that

A modular inverse only exists when a and N are *coprime*.

Example: $N = 5$

Example: $N = 4$

Modular exponentiation

Often simplified using techniques such as repeated squaring.

Example: evaluate $8^6 \bmod 11$

Method 1:

Method 2:

Method 3:

Modular arithmetic in PennyLane

PennyLane contains *built-in* operations:

- `qml.Adder`: $|x\rangle \rightarrow |x + y \bmod N\rangle$
- `qml.OutAdder`: $|x\rangle|y\rangle|b\rangle \rightarrow |x\rangle|y\rangle|b + x + y \bmod N\rangle$
- `qml.Multiplier`: $|x\rangle \rightarrow |xy \bmod N\rangle$
- `qml.OutMultiplier`: $|x\rangle|y\rangle|b\rangle \rightarrow |x\rangle|y\rangle|b + xy \bmod N\rangle$
- `qml.ModExp`: $|x\rangle|y\rangle \rightarrow |x\rangle|y \cdot a^x \bmod N\rangle$

These implementations actually leverage the QFT under the hood; see explanation in [this PennyLane demo](https://pennylane.ai/qml/demos/tutorial_shors_algorithm_catalyst#appendix-fun-with-fourier-transforms)*.

*https://pennylane.ai/qml/demos/tutorial_shors_algorithm_catalyst#appendix-fun-with-fourier-transforms

Order finding

Input: integers a and N , where $N \geq 2$ and $\gcd(a, N) = 1$

Output: smallest integer r for which $a^r \equiv 1 \pmod{N}$

Exercise: find the order of $a = 5$ for $N = 7$.

Order finding vs. period finding

Consider the function

Order finding is equivalent to *period finding* for $f_{N,a}(x)$:

Order finding on a quantum computer

Define a unitary operation that performs

If r is the order of a modulo N , and we apply $U_{N,a}$ r times,

Order finding on a quantum computer

$U_{N,a}$ has the same order, r . We can use QPE to efficiently find r with a clever initialization of the target register.

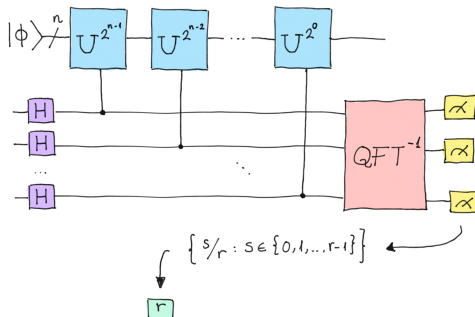


Image credit: Xanadu Quantum Codebook node S.3

Order finding on a quantum computer

Suppose there exists a state $|\phi\rangle$ for which

Consider the state

Apply U to $|\Psi_0\rangle$:

Order finding on a quantum computer

Now consider the state

Apply U to $|\Psi_1\rangle$:

Order finding on a quantum computer

This generalizes to $|\Psi_s\rangle$

It has eigenvalue

Idea: create *any* $|\Psi_s\rangle$, run QPE to get estimate of s/r , compute r .

What's the problem?

Order finding on a quantum computer

Solution: construct the uniform superposition of all of them.

But what does this equal?

Order finding on a quantum computer

The superposition of all $|\psi_s\rangle$ is just our original state $|\phi\rangle$!

$$\begin{aligned}
 |\psi\rangle &= \frac{1}{\sqrt{r}} \left(|\psi_0\rangle + |\psi_1\rangle + \dots + |\psi_{r-1}\rangle \right) \\
 &= \frac{1}{\sqrt{r}} \cdot \left(\frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{\frac{-2\pi i}{r}} U|\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)}{r}} U^{r-1}|\phi\rangle \right) \right. \\
 &\quad \left. + \frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{\frac{-2\pi i(r-1)}{r}} U|\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)^2}{r}} U^{r-1}|\phi\rangle \right) \right. \\
 &\quad \left. + \dots + \frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{\frac{-2\pi i(r-1)}{r}} U|\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)^2}{r}} U^{r-1}|\phi\rangle \right) \right) \\
 &\quad \underbrace{\qquad\qquad\qquad}_{\substack{r \quad 0 \quad \dots \quad 0 \quad \dots \quad 0}} \\
 &= \frac{1}{\sqrt{r}} \cdot \frac{1}{\sqrt{r}} \cdot r |\phi\rangle = \boxed{|\phi\rangle}
 \end{aligned}$$

If we run QPE, the output will be s/r for one of these states.

Image credit: Xanadu Quantum Codebook node S.3

Next time

Content:

- Factoring with Shor's algorithm

Action items:

- ① Tutorial Assignment 3 due Friday 23:59
- ② Midterm checkpoint report due Friday **5:00pm**

Recommended reading:

- Codebook modules QFT, QPE, SH
- Nielsen & Chuang 5.3, Appendix A.5
- PennyLane demo: Quantum just-in-time compiling Shor's algorithm with Catalyst